

FORECASTING STUDY

Forecasting Market Volatility: Where a Simple Model Helps, and Where It Quietly Breaks

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Can historical VIX data reliably predict where volatility is heading? This study puts three forecasting approaches head-to-head: a simple “do-nothing” baseline, a classical ARIMA model, and Meta’s Prophet, tested across 36 monthly experiments on data from 2023 to 2026 at one-day, one-week, and one-month horizons. ARIMA edges out the baseline by about 7% at a one-day horizon, calling the right direction 69% of the time. Stretch that to one month, and the edge disappears: the model is no better than a coin flip. What turns out to be more instructive than the error numbers is what the mistakes look like. They are not random. The errors cluster around real market events, run far larger than any well-behaved model would predict, and compound during turbulent periods. That is not a flaw to paper over; it is a precise account of what a linear model can and cannot do with volatility data, and a starting point for building something better.

This is the first of five data science projects building toward a live market-prediction dashboard. Beginning with volatility forecasting rather than something flashier is a deliberate choice: at this stage, the discipline of evaluating a model honestly matters more than how sophisticated it looks.

Most forecasting write-ups report a single error number and stop. This one keeps going, because the interesting question is not whether ARIMA can beat a naive baseline. It is where it stops being trustworthy, and why.

1. The CBOE Volatility Index

The VIX reflects 30-day implied volatility of S&P500 options. In calm markets it tends to sit in the high teens; during genuine stress it can double or triple in a matter of days. Figure 1 covers the full history from 1990 to 2026.

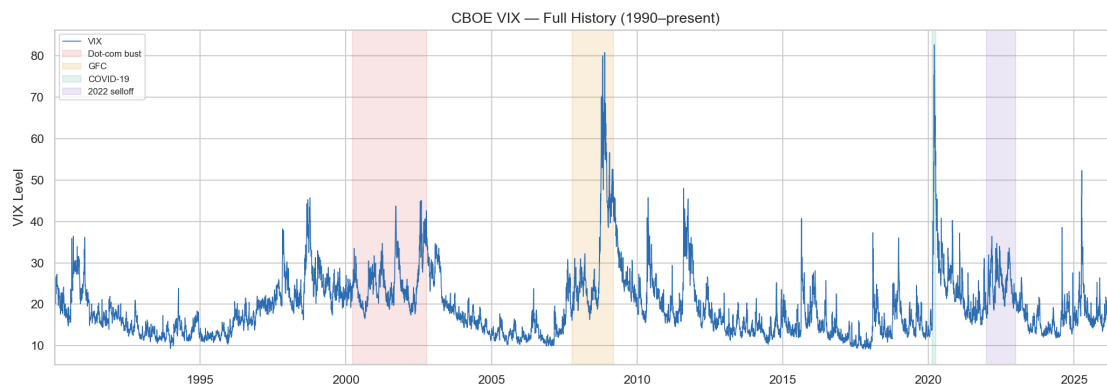


Figure 1. Full VIX history (1990–2026). Four major spike regimes: dot-com bust, Global Financial Crisis, COVID-19 onset (March 2020), and the 2022 equity selloff.

Two unit-root tests agree that VIX levels are stationary. The Augmented Dickey-Fuller test rejects the presence of a unit root ($p \approx 0$), while KPSS finds no evidence against stationarity ($p \approx 0.10$). No differencing is needed. Figure 2 shows the autocorrelation structure: the decay is slow, consistent with an AR(1) coefficient of around 0.984. The series is highly persistent, but it does not drift.

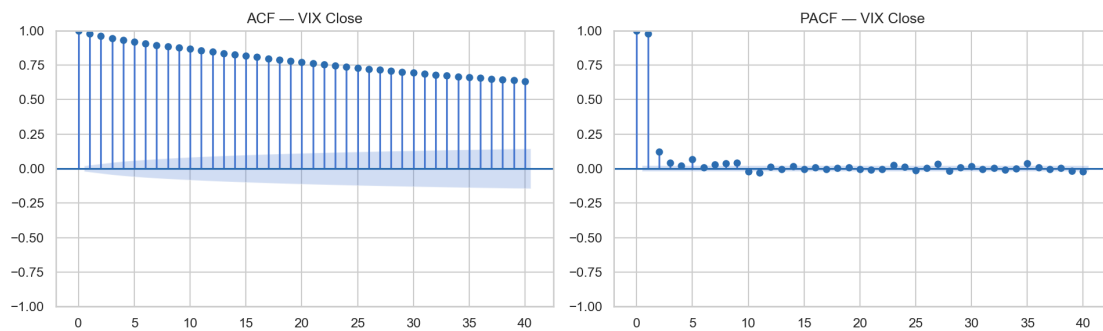


Figure 2. ACF and PACF of VIX levels. Slow ACF decay is consistent with $AR(1) \approx 0.984$ persistence; PACF cuts off at lag 2.

2. Model Selection

Naive persistence simply carries today's value forward as tomorrow's forecast. It is the baseline every other model must beat to justify its existence.

ARIMA(1,0,2) was chosen by AIC over a grid of candidate specifications fitted to the full 36-year history. Two alternatives were explicitly tested and set aside: a seasonal term at the weekly lag turned out to be essentially zero (coefficient 0.0005, $p=0.923$), and fitting on log-VIX instead of raw levels moved the error metrics by less than 2%. The level-scale ARIMA without seasonal terms is what the data supports.

Prophet was included not as a serious competitor but as a test of a specific question: does VIX have weekly or annual seasonality? Its Fourier components were built for exactly this kind of structure. The results say no.

3. Walk-Forward Validation

Rather than a single train/test split, each model is re-estimated monthly using all data available up to that point, then asked to forecast one day, one week, and one month ahead on data it has never seen. The window advances 21 trading days at a time across three years of test data, producing 36 independent forecast exercises. Every performance number reported reflects how the model would actually behave in use.

4. Results

ARIMA(1,0,2) comes out ahead at every horizon on every metric (Figures 3 and 4). At one day out, the average error is about 1.09 VIX points, roughly 7% below the naive baseline, and the model calls the right direction 69% of the time. At five days the directional edge narrows to 56%, and at one month it reaches 50.4%: statistically indistinguishable from guessing.

Prophet's performance at point forecasting is poor across the board, with RMSE running about five times higher than ARIMA at every horizon. Its additive trend component consistently pulls forecasts upward in low-volatility periods, chasing a long-run level that does not exist in a mean-reverting series. The weekly Fourier terms tell a similar story: VIX does have a small day-of-week pattern (about 0.44 points peak-to-trough), but Prophet amplifies it by a factor of nine. Its directional accuracy, oddly, stays close to ARIMA's throughout. Getting the magnitude right and getting the direction right are two separate problems.

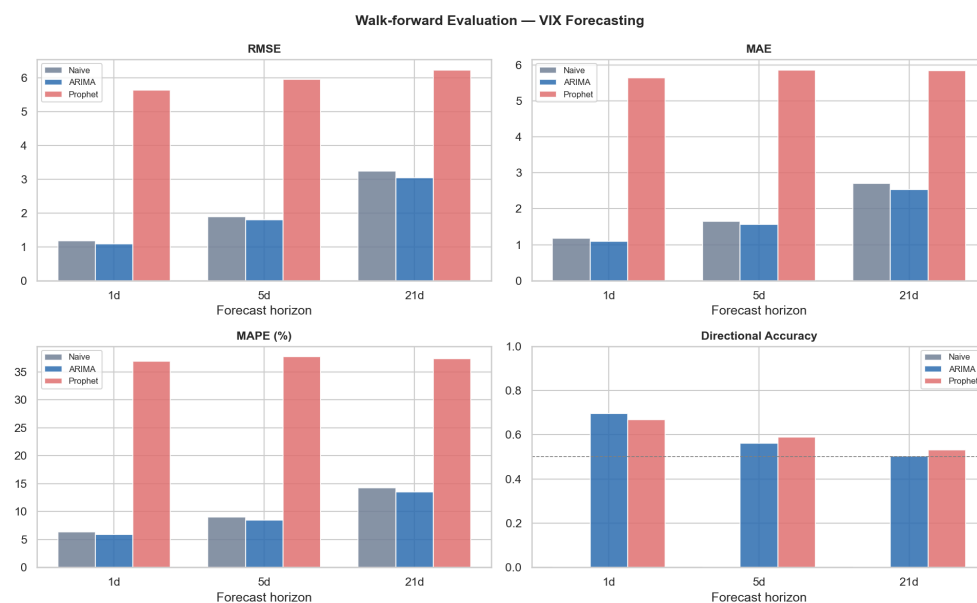


Figure 3. Walk-forward metrics across all models and horizons. ARIMA(1,0,2) leads on all error metrics; Prophet's RMSE is $\sim 5\times$ larger.

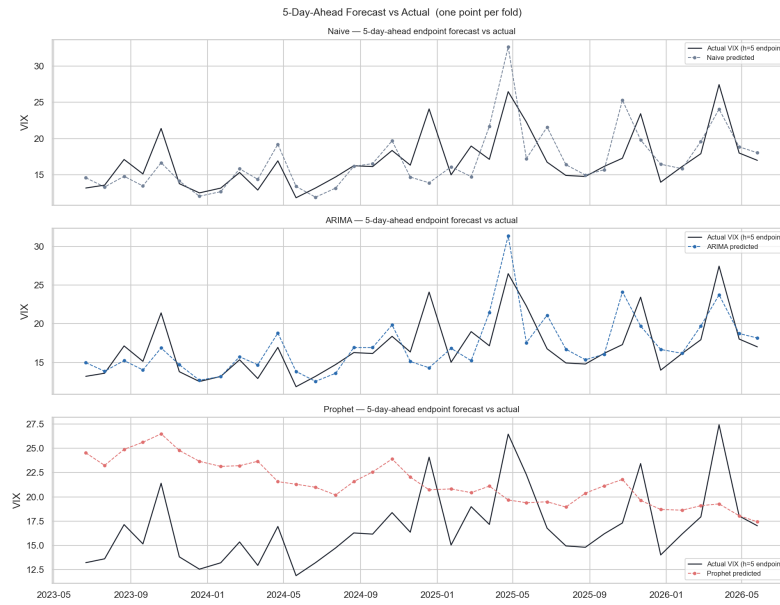


Figure 4. $h=1$ walk-forward forecasts vs. actual VIX, 2023–2026 test window. Prophet forecasts are systematically elevated.

5. Residual Diagnostics

Three standard diagnostic tests were run on the residuals of ARIMA(1,0,2) fitted to the full history of 9,183 daily observations. All three reject their null hypotheses at $p < 0.001$ (Table 1).

Test	Statistic	H_0 rejected
Ljung-Box Q(20)	91.07	Residuals are white noise
Jarque-Bera	315,931	Residuals are Gaussian
ARCH-LM (lag=12)	1,571.17	No variance clustering

Table 1. Residual diagnostic tests on ARIMA(1,0,2) fitted to the full history ($n = 9,183$). All reject at $p < 0.001$.

The residuals are fat-tailed (excess kurtosis of 31.4, skewness of 2.23) and their variance is not constant: calm stretches produce small errors, and volatile stretches produce runs of large ones. None of this disqualifies the point forecasts, but it does mean the model's confidence intervals should not be taken seriously.

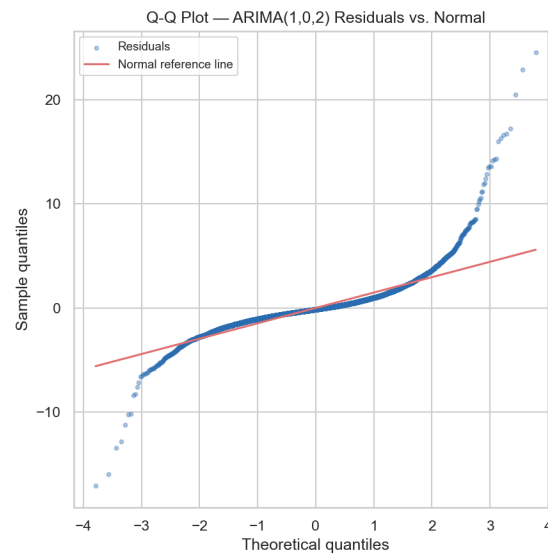


Figure 5. Q-Q plot of ARIMA(1,0,2) residuals vs. the normal distribution. Heavy tails in both directions confirm excess kurtosis of 31.4.

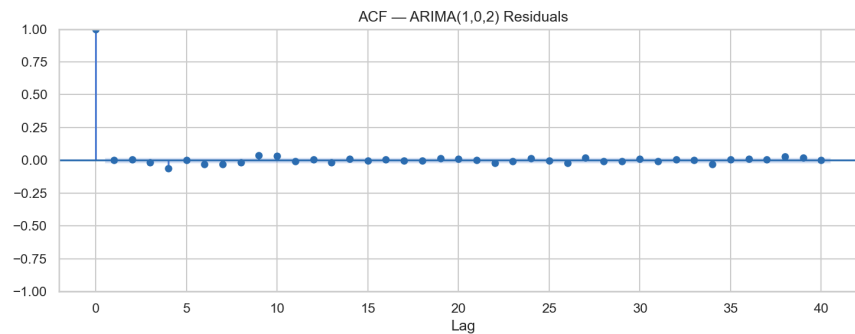


Figure 6. ACF of ARIMA(1,0,2) residuals. Several lags breach the 95% confidence band, consistent with Ljung-Box $Q(20) = 91.07$.

6. Discussion

What the diagnostics add up to is a fairly clear agenda for what comes next. Persistent variance clustering is a textbook case for a GARCH model layered on top of the ARIMA residuals, which would give the variance its own equation rather than treating it as constant. Spike prediction is a harder problem. VIX shocks are triggered by external events (rate decisions, credit blowups, geopolitical crises) that leave no fingerprint in historical VIX values themselves. Getting ahead of those moves would require bringing in outside data: credit spreads, options-positioning indicators, macro surprise indices. That is a different project.

The evaluation harness built here is model-agnostic and carries forward into the rest of this portfolio: a momentum strategy backtest, a Monte Carlo risk model, and eventually a live forecasting dashboard. The real output of this study is not an ARIMA model. It is a documented, testable account of where the limits of linear forecasting sit for this particular series.

7. Conclusions

Of the three approaches tested, ARIMA(1,0,2) is the clear winner, consistently outperforming the naive baseline by around 7% on RMSE across all 36 monthly test folds. At a one-day horizon it gets the direction right roughly two thirds of the time; at one month, it is no better than a coin flip. The residual analysis tells the rest of the story: the errors are not random noise but structured departures driven by volatility spikes, clustered in time, and far heavier-tailed than a Gaussian model assumes. Taken together, these results define exactly where a linear approach reaches its ceiling with VIX, and what a more complete model would need to address.